

INDEX

SNO	CHAPTER NAME	LAQ'S	PgNo
1.	ATOMIC STRUCTURE	L ₁ , L ₂	1 to 4
2.	PER PERIODIC CLASSIFICATION	L ₃ , L ₄ , L ₅	4 to 8
3.	CHEMICAL BONDING	L ₅ , L ₆ , L ₇	7 to 11
4.	SAQ'S		
1.	CHEMICAL BONDING	S ₁ , S ₂ , S ₃ , S ₄ , S ₅ , S ₆	12 → 16
2.	STATES OF MATTER	S ₇ , S ₈ , S ₉ , S ₁₀ , S ₁₁	16 - 20
3.	STIOCHIOMETRY	S ₁₂ , S ₁₃ , S ₁₄ , S ₁₅	20 - 23
4.	THERMO DYNOUNICS	S ₁₅ , S ₁₆ , S ₁₇ , S ₁₈	23 - 25
5.	H ₂ AND ITS COMPOUNDS	S ₁₉ , S ₂₀ , S ₂₁ , S ₂₃	25 - 28
6.	GROUP - 14	S ₂₄ , S ₂₅	28
7.	CHEMICAL EQUILIBRIUM	S ₂₆ , S ₂₇ , S ₂₈	29 - 32
	VSAQ'S		
1.	CHEMICAL BONDING	V ₁ , V ₂ , V ₃ , V ₄ , V ₅	32 -
2.	STATES OF MATTER	V ₆ , V ₇ , V ₈ , V ₉	33
3.	STIOCHIOMETRY	V ₁₀ , V ₁₁ , V ₁₂ , V ₁₃ , V ₁₄ , V ₁₅ , V ₁₆	34 - 36 38
4.	THERMO DYNOUNICS	V ₁₇ , V ₁₈	36
5.	GROUP - 1 AND 2 H ₂ AND ITS COMPOUNDS	V ₁₉ , V ₂₀ , V ₂₁ , V ₂₂ , V ₂₃ , V ₂₄ , V ₂₅ , V ₂₆ , V ₂₇	36 - 38
6.	CHEMICAL EQUILIBRIUM	V ₂₈ , V ₂₉ , V ₃₀ , V ₃₁ , V ₃₂	38
7.	HYDROGEN AND ITS COMPOUNDS	V ₃₃ , V ₃₄ , V ₃₅	39

8.

GROUP - 13

 V_{36}, V_{37}, V_{38}

39

TOTAL LAQ'S - 7

TOTAL SAQ'S - 28

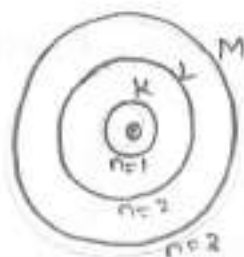
TOTAL VSQ'S - 38

①

QAQ-1

Q) What are the postulates of Bohr's model of hydrogen atom? Discuss the importance of this model to explain various series of line spectra in hydrogen atom?

- A) 1) The electron in hydrogen atom revolves around the nucleus in a fixed circular path called orbits. (4M)
- 2) Each orbit is associated with definite energy. so these are called energy levels represented as 1, 2, 3, 4... or K, L, M, N...



3) As long as the electron revolves in an orbit neither loses nor gains energy. so the energy of an electron in an orbit remains constant. these are called stationary orbits.

4) The angular momentum is given by $mvr = \frac{nh}{2\pi}$

5) If an electron jumps from one level to another level it emits energy or absorbs energy.

$$\Delta E = E_2 - E_1$$

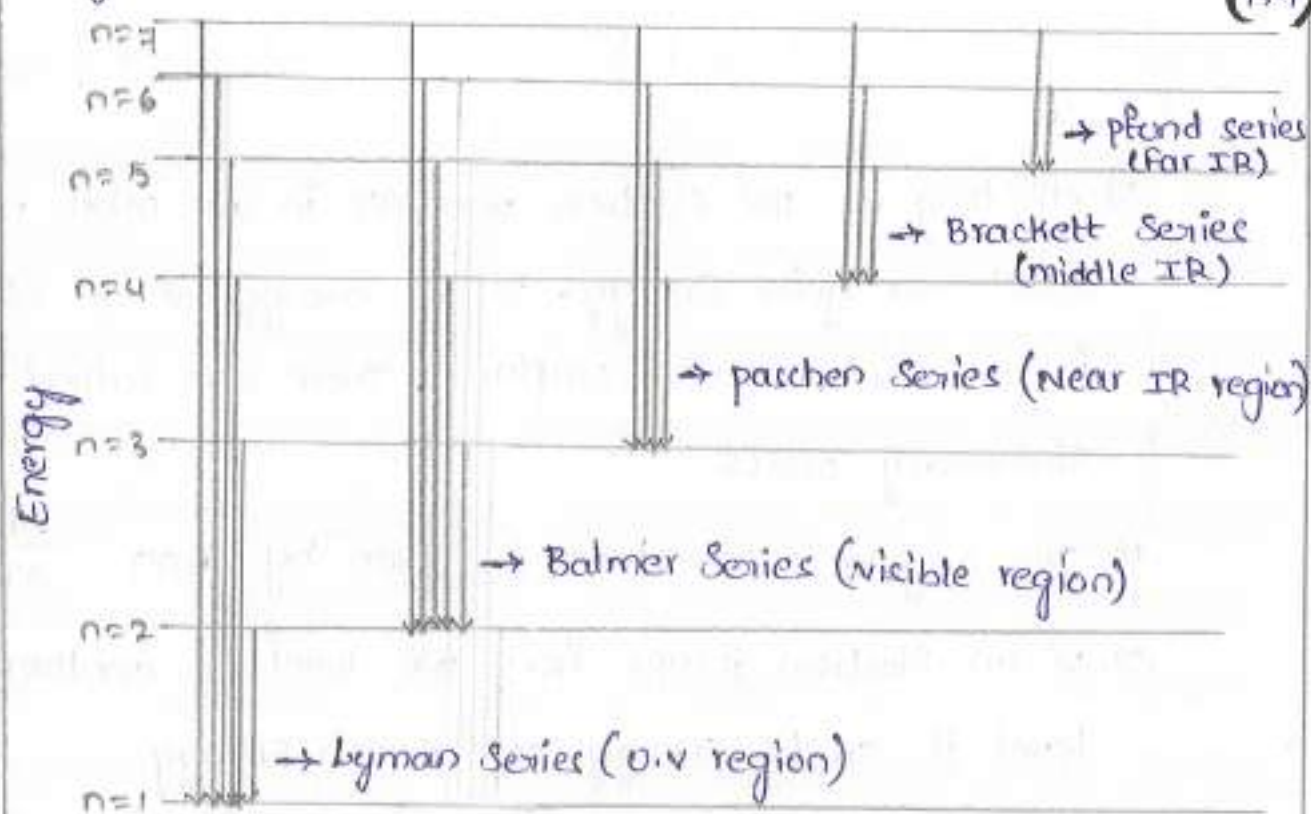
6) If electron jumps from lower orbit to higher orbit it absorbs energy.

(2)

1. If Electron jumps from higher orbit to lower orbit then it emits Energy.

Bohr's Explanation to hydrogen spectrum: (2 M)

- 1) Hydrogen atom has one proton and one Electron, Electron jumps to higher Energy level which is unstable. Hence it comes back to ground state by emitting the Energy.
- 2) Since a sample of hydrogen contains many atoms, several spectral lines are formed. They are Lyman, Balmer, Paschen, Brackett and Pfund series. (1 M)



Rydberg's Equation

$$\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$R = 109677 \text{ cm}^{-1}$$

(1 M)

(3)

(L₂) How are the quantum numbers n, l and m, s arrived at? Explain the significance of these quantum no.?

A) The quantum number n, l and m are arrived from ^(2M) Schrodinger wave Equation. They are four quantum numbers.

1) Principal quantum number:- (n)

(1/2 M)

i. It was proposed by Neils Bohr.

ii. The total number of subshells for a given value of $n = n$

iii. The maximum number of electrons for a given value of $n = 2n^2$

iv. Significance:- It indicates the size of an orbit and Energy of an electron.

2) Orbital angular momentum:- (l)

(1/2 M)

i. It was proposed by Sommerfeld.

ii. The value of l depends on ' n '. The value of ' l ' are from 0 to $(n-1)$

iii. When l values are 0, 1, 2, 3, ... the subshells are called s, p, d, f, ... respectively

Significance:- It indicates the shape of orbital.

3) Magnetic quantum number:- (m)

(1/2 M)

i. It was proposed by Lande

(4)

(4)

ii, The values of 'm' depends on 'l'. The values of m are from -l to +l including '0'.

iii, The number of Electrons in a subshell = $2(2l+1)$

Significance:- It indicates the orientation of orbital in space.

4) Spin quantum number:- (s) (1/2 M)

i, It was proposed by Goudsmith and uhlenbeck.

ii, It is independent. The values of s are $+\frac{1}{2}$ and $-\frac{1}{2}$

iii, The maximum no. of Electrons in a orbital = 2

Significance:- It indicates the direction of spin of electron.

"

(L3) Define IE_1 and IE_2 . Why is $IE_2 > IE_1$ for a given atom? Discuss the factors that effect IE of an element?

A) Ionization Enthalpy:- The minimum amount of (2M)
Energy required to remove the most loosely bound electron from a neutral, isolated gaseous atom is called first ionization enthalpy.

First ionization Enthalpy:- The Energy required to remove an electron from a neutral, isolated gaseous atom is called first ionization enthalpy.

(5)

Second ionization Enthalpy: The energy required to 2M remove an electron from unipositive gaseous ion is called second ionization Enthalpy.

Second ionization Enthalpy is always more than the first ionization Enthalpy.

It is due to increase of effective nuclear charge from neutral atom to unipositive ion.

Factors influencing ionization Enthalpy:

(4M)

1) Atomic radius:

Ionization Enthalpy $\propto \frac{1}{\text{atomic radius}}$

2) Nuclear charge:

Ionization Enthalpy $\propto$ Nuclear charge.

3) Screening effect:

Ionization Enthalpy $\propto \frac{1}{\text{Screening effect}}$

4) penetration power:

Ionization Enthalpy $\propto$ penetration power.

5) Half filled (or) completely filled subshells:

→ High ionization Enthalpy

⑥

(4) What is a periodic property? How the following properties vary in a group & in a period? Explain

a) Atomic radius b) IE c) EN

d) Electron gain Enthalpy (e) electron affinity

e) Nature of oxides.

(2M)

A) periodic properties:- The repetition properties of element at certain regular intervals of electronic configuration.

periodic property	period	Reason	group	Reason
Atomic radius	↓	↑ (N.A)	↑	↓ (N.A)
Ionization Enthalpy	↑	↓ (A.S)	↓	↑ (A.S)
Electron Negetivity	↑	↓ (A.S)	↓	↑ (A.S)
Electron affinity	↑	↓ (A.S)	↓	↑ (A.S)
Nature of Acidic	↑	-	↓	-
oxides in Basic	↓	-	↑	-

N.A → Nuclear attraction ↑ → Increases

A.S → Atomic size ↓ → decreases

Atomic radius:-

In period! left to right decreases

Reason! Nuclear attraction increases.

In group! Top to bottom increases

Reason! Nuclear attraction decreases.

Ionisation Enthalpy:-

In period! left to right increases.

Reason! Atomic size decreases.

In group! Top to bottom decreases.

Reason! Atomic size increases.

Electro negativity:-

In period! left to right increases

Reason! Atomic size decreases

In group! Top to bottom decreases

Reason! Atomic size increases.

Electron Affinity:-

In period! left to right increases

Reason! Atomic size decreases

In group! Top to bottom decreases

Reason! Atomic size increases.

Nature of acidic:-

In period! left to right increases

In group! In group! Top to bottom decreases.

Oxides in Base:-

In period! left to right decreases.

In group! Top to bottom increases.

(b5) Write an essay on s, p, d and f block elements.

A) Based on the presence of differentiating electron, the elements are classified into four blocks.

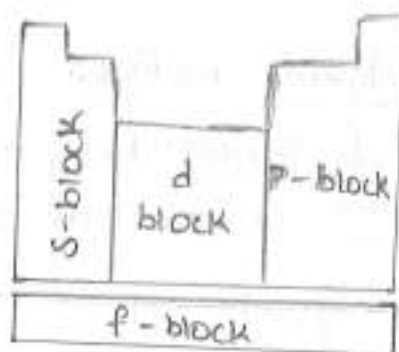
1) s-block elements:

2M

i. The elements in which differentiating electron enters into s-orbital is called s-block elements.

ii. The general electronic configuration is ns^{1-2}

iii. It contains two groups (i.e) I A and II A.

2) p-block elements:

(2M)

i. The elements in which differentiating electron enters into p orbital are called p-block elements.

ii. The general electronic configuration is $ns^2 np^{1-6}$.

iii. It contains 6 groups from 13 to 18.

3) d-block elements:

(2M)

i. The elements in which differentiating electron enters into $(n-1)d$ subshell are called d-block elements.

ii. The general electronic configuration is $(n-1)d^{1-10} ns^{1-2}$

iii. It contains 8 to 12 groups.

4) f-block elements:

(2M)

i. The elements in which differentiating electron enters into $(n-2)f$ subshells are called f-block elements.

ii. The E.C is $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$

iii. It contains two series (ie) lanthanides and actinides.

(b6) What do you understand by hybridisation? Explain different types of hybridisation involving s & p orbitals?

A) Hybridisation: The inter mixing of atomic orbitals to form new, same number of equivalent orbitals.

(9)

is called hybridisation.

Types of Hybridisation:-

1) Sp - hybridisation:-

(2M)

- i. The mixing of one s orbital and one p orbital to give 2 sp hybrid orbital is called sp hybridisation.
- ii. Shape is linear and bond angle is 180° .
- iii. s character is 50% & p character is 50%.

Ex:- formation of BeCl_2

2) Sp² - hybridisation:-

(2M)

- i. The mixing of one s orbital and two p orbitals to give three sp² hybrid orbitals is called sp² hybridisation.
- ii. Shape is trigonal planar & bond angle is 120° .
- iii. s character is 33.3% & p character is 66.6%.

Ex:- formation of BCl_3

3) Sp³ - hybridisation:-

(2M)

- i. Inter mixing of one s orbital and three p orbitals to give four sp³ hybrid orbitals is called sp³ hybridisation.
- ii. Shape is tetrahedral and bond angle is $109^\circ 28'$.

iii, s character is 25% & p character is 75%.

Ex: formation of CH_4

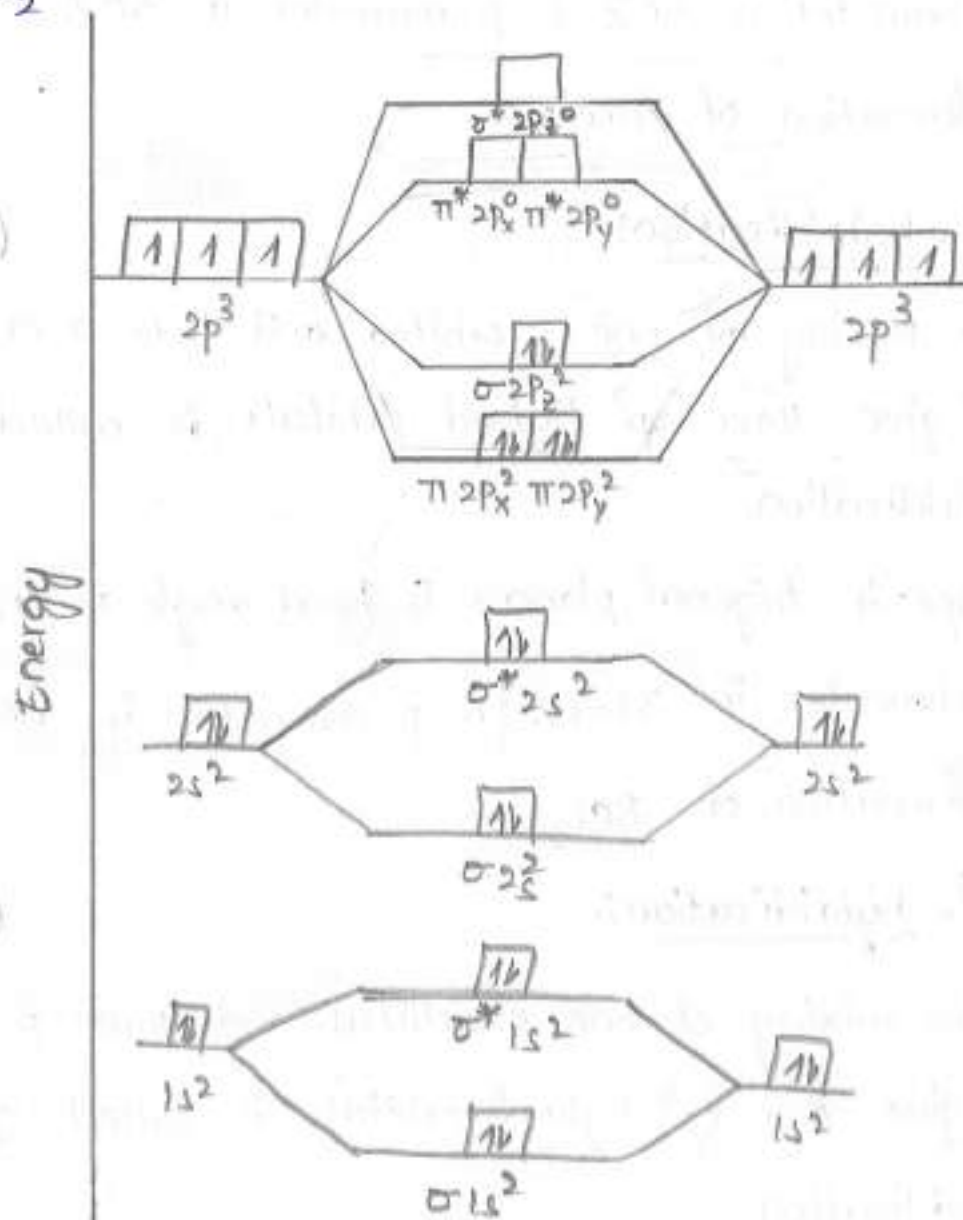
(pg. 19)
(L-1)

Give the molecular orbital energy diagram of N_2 and O_2 . calculate the respective bond order.

Write the magnetic nature of N_2 and O_2 molecules.

(4M)

A) a) N_2



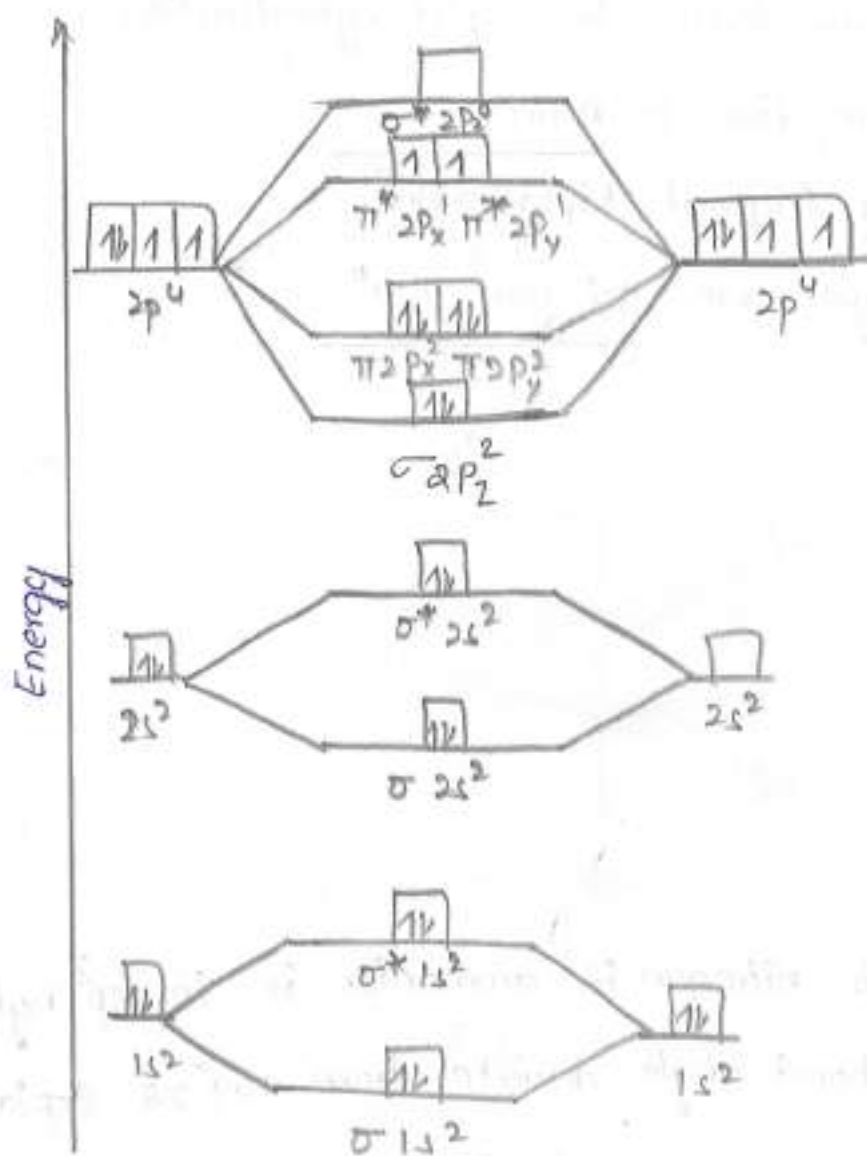
(11)

$$\begin{aligned} \text{Bond order} &= \frac{10-4}{2} \\ &= 6/2 \\ &= 3 \end{aligned}$$

→ It is diamagnetic in nature.

(b) O_2

(4M)



$$\begin{aligned} \text{Bond order} &= \frac{10-6}{2} \\ &= 4/2 = 2 \end{aligned}$$

→ It is para magnetic in nature.

SAQ's :-

(S₁) Explain the hybridisation involved in PCl_5 molecule. (4M)

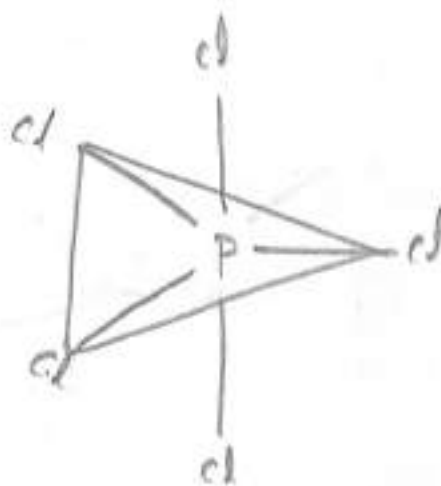
A) i. In PCl_5 , P is the central atom.

ii. The inter mixing of one s and three p, one d orbitals of P atom forms five sp^3d hybridisation.

iii. It forms five σ bonds.

iv. Shape is trigonal bipyramidal.

v. Bond angles are 90° and 120° .



(S₂) Even though nitrogen in ammonia is in sp^3 hybridisation, the bond angle deviate from $109^\circ 28'$ Explain. (2M)

A) The change in bond angle in ammonia from $109^\circ 28'$ to 107° can be explained by VSEPR Theory.

i. In NH_3 , central atom is N. It contains one lone pair

and three bond pairs.

(2M)

ii, sp^3 hybridisation, tetrahedral, bond angle $109^\circ 28'$

iii, According to VSEPR theory repulsion between lone pair and bond pair causes the decrease in bond angle to 104° and shape changes to pyramidal.

(S₃) Explain the hybridisation involved in SF_6 molecule.

(4M)

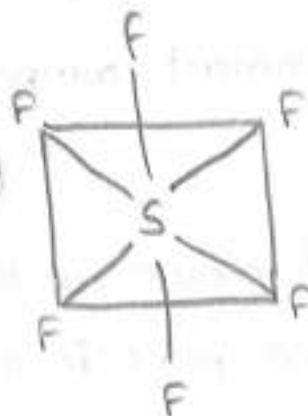
A) 1) In SF_6 molecule, S is central atom.

2) The inter mixing of one 's' and three 'p' and two 'd' orbitals of 'S' atom forms six sp^3d^2 hybrid orbitals.

3) It forms six σ bonds.

4) Shape is octahedral

5) Bond angles are 90° and 180°



(S₄) Explain Fajan's rule and give suitable examples.

(4M)

A) 1) For a given cation covalent character increases with increasing size of anion.

Ex: KI is more covalent than KF

- 2) covalent character increases with increasing charges either cation or anion.

Ex: SnCl_4 is more covalent than SnCl_2

- 3) for a given anion covalent character increases with decrease in size of cation.

Ex: LiF is more covalent than KF

- 4) covalent character is higher for compounds having cations with pseudo inert gas than with inert gas configuration.

Ex: CuCl is more covalent than NaCl .

"

(S₅) Define Dipole moment. Write its applications.

- a) NH_3 has a higher dipole moment compared to NF_3 . Why? (2M)

- A) The product of magnitude of charge on the poles and the distance between two poles is called dipole moment.

$$\mu = q \times d$$

Applications:

(2M)

- i. Geometry of molecule can be determined
- ii. percentage of ionic character can be calculated.

iii) cis and trans isomers can be separated.

iv) Bond angle can be determined.

a)

A) 1) In case of ammonia due to lone pair on N is the same direction as the resultant dipole moment of N-H bonds.

2) In NF_3 , the orbital dipole moment is direction opposite to resultant dipole moment.

(b) Dipole moment observed for NF_3 and not BF_3 why?

A) i) BF_3 is non-polar molecule with polar bonds.

ii) Net dipole moment is zero.

iii) In NF_3 is polar molecule with polar bonds, Net dipole is positive.

(S6) What is hydrogen bond Explain the different types of hydrogen bonds with examples. (2M)

A) The weak electrostatic force of attraction between a partially positively charged hydrogen atom and a highly electro negative atoms like N, O, F of the same (or) different is known as hydrogen bond. There are two types

i, inter molecule hydrogen bond in the (1M)

The hydrogen bond is formed between two different polar molecules.

Ex: $\text{H}-\text{F} \cdots \cdots \text{H}-\text{F}$

ii, Intramolecule hydrogen bonding: (1M)

The hydrogen bond is formed between two atoms of same molecule.

Ex: o-nitrophenol.

(S₇) Write the postulates of kinetic molecular theory of Gases. (4M)

- A)
- i, Gases consists large number of tiny particles called molecules.
 - ii, Gas molecules move randomly in all directions with high speeds.
 - iii, Molecules motions are unaffected by gravity.
 - iv, collisions of gas molecules are perfectly elastic.
 - v, pressure is exerted by the gas as result of collision of the particles with the walls of the container.
 - vi, Average kinetic energy of gas molecules is directly

(17)

proportional to the absolute temperature $KE \propto T$

(S8) Deduce (a) Boyle's law (b) Charles's law

(c) Graham's law (d) Dalton's law (e) Kinetic Energy from Kinetic gas equation.

(2M)

A) (a) Boyle's law:- According to Kinetic gas equation

$$PV = \frac{1}{3} mn\bar{u}_{rms}^2$$

$$PV = \frac{2}{3} \cdot \frac{1}{2} mn\bar{u}^2 \rightarrow (1)$$

$$\frac{1}{2} mn\bar{u}^2 = KE \rightarrow (2)$$

$$K.E = KT \rightarrow (3)$$

from Equ (1), (2) and (3)

$$PV = \frac{2}{3} KT \quad \therefore PV = \text{constant}$$

(2M)

(b) Charles's law:- from Kinetic gas equation.

$$PV = \frac{1}{3} mn\bar{u}_{rms}^2$$

$$PV = \frac{2}{3} \cdot \frac{1}{2} mn\bar{u}^2 \rightarrow (1)$$

$$PV = \frac{2}{3} KE$$

$$PV = \frac{2}{3} KT$$

$$V = \frac{2}{3} \frac{KT}{P}$$

$$\therefore V \propto T$$

(c) Graham's law: According to kinetic gas equation

$$PV = \frac{1}{3} mn\bar{u}^2$$

$$PV = \frac{1}{3} M\bar{u}^2$$

$$\left[\bar{u} = \sqrt{\frac{3PV}{M}} \right]$$

$$u_{rms} = \sqrt{\frac{3P}{d}}$$

$$\frac{u}{M} = \frac{1}{d}$$

$$u_{rms} \propto \frac{1}{\sqrt{d}}$$

$$\therefore r \propto \frac{1}{\sqrt{d}}$$

(d) Dalton's law:

$$P_1 = \frac{1}{3} \frac{m_1 n_1 \bar{u}_1^2}{V} \rightarrow (1)$$

$$P_2 = \frac{1}{3} \frac{m_2 n_2 \bar{u}_2^2}{V} \rightarrow (2)$$

$$P_{total} = \frac{1}{3} \frac{m_1 n_1 \bar{u}_1^2}{V} + \frac{1}{3} \frac{m_2 n_2 \bar{u}_2^2}{V}$$

$$P = P_1 + P_2$$

(Q) State and explain Graham's law of Diffusion? (2M)

A) At constant temperature and pressure rate of diffusion of given mass of gas is inversely proportional to square root of its density.

$$r \propto \frac{1}{\sqrt{d}}$$

Molar mass is directly proportional to density ($M \propto d$)

Molar mass = 2 x vapour density. ($M \propto VD$)

(19)

If r_1, r_2 are the rates of diffusion of two different gases then Graham's law. (2M)

$$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{V D_2}{V D_1}}$$

$$\frac{r_1}{r_2} = \frac{v_1 t_2}{v_2 t_1}$$

(S10) State and explain Dalton's law of partial pressure?

A) At constant temperature and volume, the total (2M)
pressure exerts by a mixture of non reacting gases
is equal to the sum of the partial pressure of all
individual gases

$$P = P_1 + P_2 + P_3 + \dots \dots \dots (2M)$$

According to Dalton's law.

$$= \frac{n_1 RT}{V} + \frac{n_2 RT}{V} + \frac{n_3 RT}{V}$$

$$= (n_1 + n_2 + n_3) \frac{RT}{V}$$

$$P = \frac{nRT}{V}$$

$$\frac{P_1}{P} = \frac{\frac{n_1 RT}{V}}{\frac{nRT}{V}} = \frac{n_1}{n} = x_1$$

$$\boxed{P_1 = x_1 P}$$

(S11) Derive an ideal gas equation?

A) According to Boyle's law $V \propto \frac{1}{P}$ (4M)

According to charle's law $V \propto T$

According to Avagadro's law $V \propto n$

$$\therefore V \propto \frac{nT}{P}$$

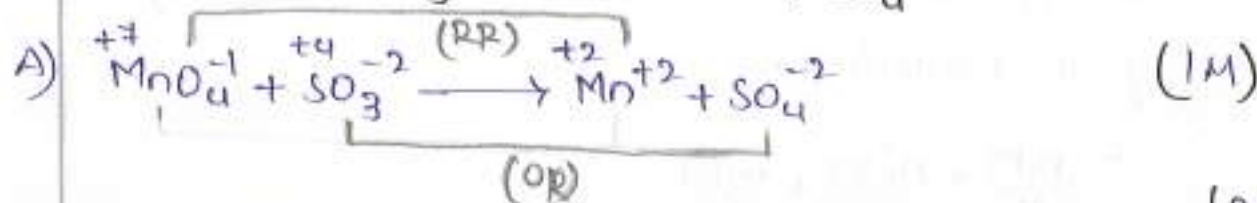
$$PV \propto nT$$

$$PV = nRT$$

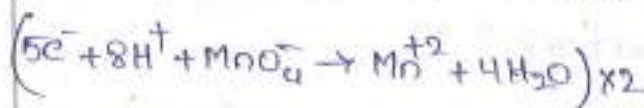
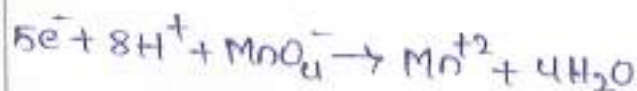
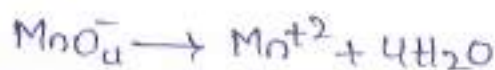
$$R = \frac{PV}{nT}$$

$$\therefore R = 8.314 \text{ J K}^{-1} \text{ Mol}^{-1}$$

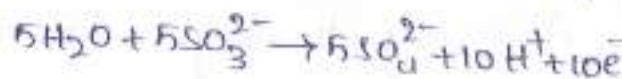
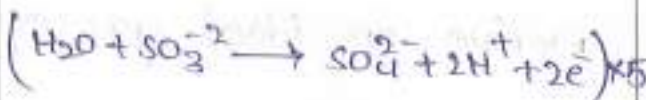
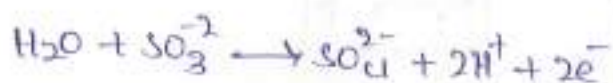
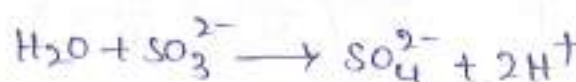
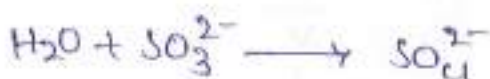
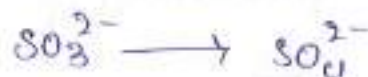
(Q12) Balance the following redox equation by ion-electron method taking place in acidic medium.



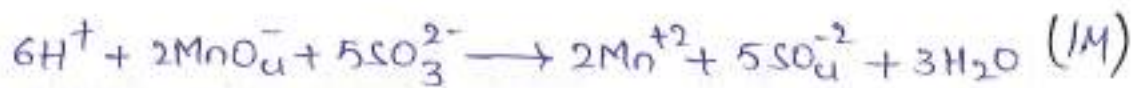
RHR



OHR



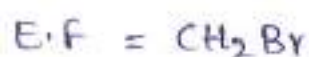
(21)



(S₁₃) Determination of Empirical formula and molecular formula.

(a) A carbon compound contains 12.8% carbon, 2.1% hydrogen, 85.1% bromine. The molecular weight of the compound is 187.9. Calculate the molecular formula. (3M)

Sym of Element	% of composition	Atomic weight	Atomic ratio = % / At. wt	Simple Ratio	Whole number ratio
C	12.8	12	$\frac{12.8}{12} = 1.06$	$\frac{1.06}{1.06} = 1$	1
H	2.1	1	$\frac{2.1}{1} = 2.1$	$\frac{2.10}{1.06} = 2$	2
Br	85.1	80	$\frac{85.1}{80} = 1.06$	$\frac{1.06}{1.06} = 1$	1



(1M)

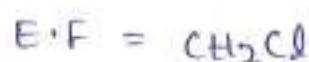
$$n = \frac{Mwt}{EFwt} = \frac{187.9}{94} = 2$$



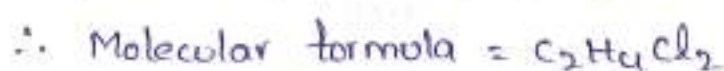
(b) A compound having 4.07% hydrogen, 24.27% carbon and 71.65% chlorine its molecular weight is 98.96. What are its empirical formula and molecular formula.

Element	% of composition	Atomic weight	Atomic ratio = % / At. Wt	Simple ratio
C	24.27	12	2.02	1
H	4.07	1	4.07	2
Cl	71.65	35.5	2.02	1

(1M)



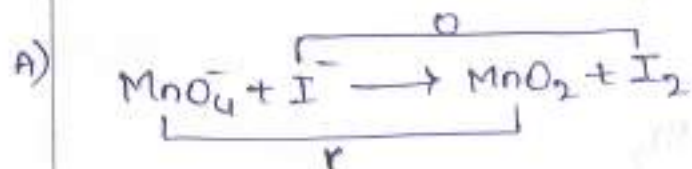
$$n = \frac{\text{Molecular Weight}}{\text{Empirical formula weight}} = 2$$



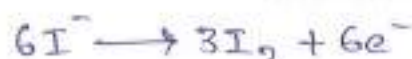
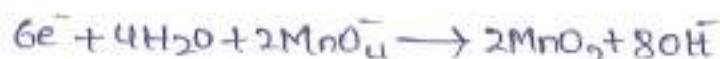
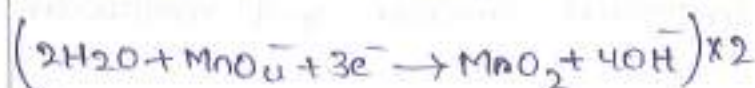
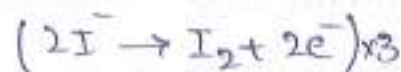
(S14) Balance the following redox equation by ion-electron method taking place in basic medium

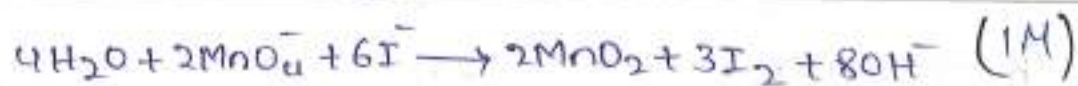


(1M)



(2M)





(S15) State the first law of thermodynamics. Explain its Mathematical notation. (2M)

A) The impossibility of construction of a perpetual motion "machine of first kind."

It is also known as law of conservation of Energy.

$$\Delta U = q + w \quad (2M)$$

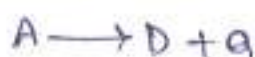
Work done in a system is positive, work done by the system is negative.

Heat observed by the system is (+ve), heat released by the system is negative.

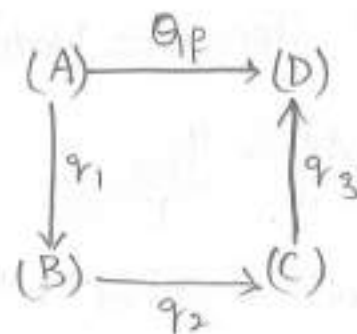
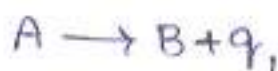
(S16) State and Explain the Hess's law of constant heat summation. (2M)

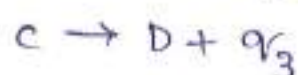
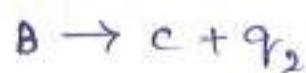
A) The total heat change in a reaction is the same wheather the chemical reaction takes place in single step or several steps.

Single step:-



Several steps:-





(2M)

$$Q = q_1 + q_2 + q_3$$

Ex: CO_2

Single step $\rightarrow \Delta H = -393.52 \text{ kJ}$

Several steps $\rightarrow \Delta H = -110.5 \text{ kJ}$

$$\Delta H = -283.02 \text{ kJ}$$

$$\text{Total } \Delta H = (-110.5 \text{ kJ}) + (-283.02) = -393.52 \text{ kJ}$$

The two ΔH values are same.

(S14) What is Entropy? Explain with Examples.

(2M)

A) The thermodynamical quantity which measures "disorder" of molecules or randomness of system is called entropy.

$\rightarrow$ It is a state function.

denoted by 'S'. & change in entropy is ΔS .

eg: $\text{Ice} \rightleftharpoons \text{Water} \rightleftharpoons \text{Vapour}$

(2M)

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

units - $\text{J mol}^{-1} \text{K}^{-1}$

(S18) State the third law of thermodynamics. What do you understand by it? (2M)

A) The Entropy of a pure and perfectly crystalline is zero at absolute zero of temperature.

$$\lim_{T \rightarrow 0} S = 0 \quad (2M)$$

When the temperature rises then Entropy increases.

Ex: $\text{Ice} \rightleftharpoons \text{Water} \rightleftharpoons \text{vapour}$.

(S19) Explain the following with suitable Examples.

(a) Electron-deficient hydrides

(b) Electron-precise hydrides

(c) Electron-rich hydrides

(d) Ionic hydrides. (1M)

A) (a) These hydrides have less number of electrons to write their Lewis structure.

Ex: B_2H_6 .

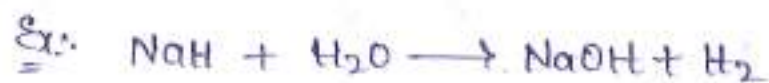
(b) These hydrides have the required number of electrons to write their Lewis structure. (1M)

Ex: CH_4

(c) These hydrides have excess electron which are present as lone pairs. (1M)

Ex: H_2O has 2 lone pairs.

(d) When hydrogen reacts with 's'-block elements (1M)
which are highly electro positive and forms ionic
hydrides.



(S₂₀) Explain the terms hard water and soft water. write
a note on the i, ion-Exchange method & ii, Calgon
method for the removal of hardness of water.

A) Soft water ∴ Water which gives good quantity of (1M)
lather with soap is soft water.

Hard water ∴ Water does not give good quantity of (1M)
lather with soap is called hard water.

(i) Hardness can be removed by ion Exchange method (1M)
the hydrated sodium Aluminium silicate is Zeo Light

→ When the hard water is passed into Zeolite
 Ca^{+2} and Mg^{+2} ions exchange Na^+ ions. (1M)

(ii) $\text{Na}_2\text{P}_2\text{O}_7$ is known as calgon when passed over
calcium and magnesium cations are removed. Hard
Water is changed to soft water.

(S₂₁) What is temporary and permanent hardness of water
Mention which ions causes the hardness of water.

- A) The hardness due to presence of bicarbonates of calcium and magnesium is called temporary hardness.
i, Boiling ii, Clark's Method. (2M)

It is due to absence of soluble salts of magnesium and calcium in the form of chlorides and sulphides in water.

i, Ion-exchange method.

ii, Calgon method.

- (S₂₂) Write a note on heavy water. (4M)

A) D_2O contains two heavy hydrogens.

→ Heavy water is prepared by exhaustive electrolysis of water containing alkali. At 4th stage D_2O is formed.

→ Heavy water has less reactivity than normal water.

M.W of D_2O = 20.

uses: 1) It is used as moderator in nuclear reactor.

2) In the preparation of deuterium.

- (S₂₃) What are electron deficient compounds? Is BCl_3 an electron deficient species. Explain. (4M)

A) These hydrides have less number of electrons to write their Lewis structure.

→ In BCl_3 , Boron is the central atom. Which is incomplete octet.

→ Boron contain 5 electrons, so it is electron deficient electron

→ BCl_3 act as Lewis acid & it is accept lone pair of electrons.

(S₂₄) Explain the difference in properties of diamond and graphite on the basis of their structure.

A) Diamond: (2M)

→ It is sp^3 hybridisation and it is tetrahedral

→ C-C bond length is $1.54 \text{ \AA}$ & bond angle is $109^\circ 28'$

→ Diamond is a hardest material.

→ Graphite: (2M)

→ It is sp^2 hybridisation and three carbon atoms to form hexagonal rings.

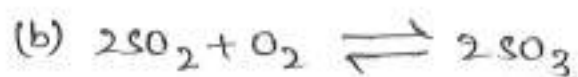
→ C-C bond length is $1.42 \text{ \AA}$ & bond angle is 120°

→ Graphite is soft lubricant

(S₂₅) Why is diamond hard? (4M)

A) It is sp^3 hybridisation and carbon atoms bound has tetrahedral. C-C bond length is $1.54 \text{ \AA}$ together held by strong covalent bonds. The amount of energy required to break this bonds is high. So it is hard.

(S₂₆) Derive the relation between K_c and K_p for equilibrium reaction. (a) $N_2 + 3H_2 \rightleftharpoons 2NH_3$



A) (a) $K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$ (1M)

WKT, $PV = nRT$

$$P = \left(\frac{n}{V}\right)RT$$

$$= CRT$$

(2M)

$$K_p = \frac{[NH_3]^2 (RT)^2}{[N_2]RT [H_2]^3 (RT)^3}$$

$$K_p = \frac{[NH_3]^2}{[N_2][H_2]^3} \cdot \frac{(RT)^2}{(RT)^4}$$

$$= \frac{[NH_3]^2}{[N_2][H_2]^3} (RT)^{-2}$$

$$K_c = K_p (RT)^{+2}$$

(1M)

$$\therefore K_p < K_c$$

(b) $K_c = \frac{[SO_3]^2}{[SO_2]^2 [O_2]}$ (1M)

$$PV = nRT$$

$$P = CRT$$

(2M)

$$\begin{aligned}
 &= \frac{[SO_3]^2 (RT)^2}{[SO_2]^2 (RT)^2 [O_2] RT} \\
 &= \frac{[SO_3]^2}{[SO_2]^2 [O_2]} \cdot \frac{(RT)^2}{(RT)^3} \\
 &= \frac{[SO_3]^2}{[SO_2]^2 [O_2]} (RT)^{-1} \quad (1M)
 \end{aligned}$$

$\therefore K_p < K_c$.

"

(S₂) Discuss the application of Lechatelier's principle for the industrial synthesis of Ammonia by Haber process.



→ It is reversible reaction

→ It is Exothermic reaction.

(3M)

Effect of concentration: With increase of concentration of reactant decreasing of concentration of products favours forward reaction.

Effect of pressure: High pressure favours reaction towards lesser volume side, hence high pressure are required for NH_3 . That is 200 atm.

Effect of temperature: Low temperature is required. so Reaction is slow,

optimum temperature :- (425 K - 475 K)

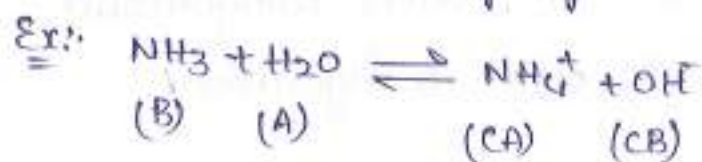
Effect of catalyst :- To speed up the reaction, iron is used as catalyst. To increase the activity of catalyst (Mo) used as promoter.

(S28) (a) What is a conjugate acid - base pair? Illustrate with Examples

(b) Write conjugate acid and bases for following species. i, H_2O ii, NH_3 iii, HCO_3^- iv, HSO_4^- v, OH^- vi, H_2O_2 .

(2M)

A) According to Bronsted-Lowry theory, the acid base pair that differ only by one proton.



→ NH_4^+ is a conjugate acid of NH_3 , OH^- conjugate base of H_2O . (2M)

(b)

Bronsted Base	conjugate acid	Bronsted acid	conjugate Base
H_2O	H_3O^+	H_2O	OH^-
NH_3	NH_4^+	NH_3	NH_2^-
HCO_3^-	$H_2CO_3^+$	HCO_3^-	CO_3^{2-}
HSO_4^-	H_2SO_4	HSO_4^-	SO_4^{2-}

(32)

(32)

	conjugate acid	conjugate base
OH^-	H_2O	O^{2-}
H_2O_2	H_3O_2^+	HO_2

* VSAQ's *

(V₁) What is octet rule? (2M)

A) The phenomenon of having 8 electrons in valency shell.

(V₂) Which of the two ions Ca^{+2} (8) Zn^{+2} is more stable & why? (2M)

A) Ca^{+2} is more stable than Zn^{+2} . octet configuration is more stable than pseudo octet configuration.

(V₃) What are the coordination numbers of NaCl & CsCl . (2M)

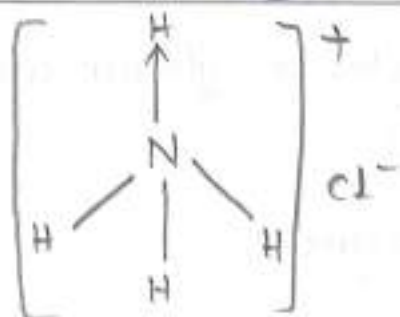
A) CN of NaCl is 6, CN of CsCl is 8.

(V₄) Why H_2O has higher Bp than HF (2M)

A) More number of hydrogen bonds are present in H_2O than HF.

(V₅) What type of bonds are present in NH_4Cl ? Write its structure. (2M)

A) covalent bond, ionic bond, dative bond.



(V6) Why the gas constant 'R' is called universal gas constant. (2M)

- A) 'R' is same for all gases. so it is universal gas constant.

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

(V7) Which of the gases diffuses faster among N_2 , O_2 and CH_4 ? Why? (2M)

- A) CH_4 diffuses faster. due to lower molar mass

$$\text{Rate of diffusion} \propto \frac{1}{\sqrt{MW}}$$

(V8) How many times methane diffuses faster than Sulphur dioxide? (2M)

- A) CH_4 diffuses two times faster than SO_2

(V9) State Boyle's law. Give its mathematical Expression. (2M)

- A) At constant temperature the volume of a gas is inversely proportional to $\frac{1}{\text{pressure}}$.

$$V \propto \frac{1}{P}$$

$$PV = K.$$

(V₁₀) How many number of moles of glucose are present in 540 gms of glucose? (2M)

A) $MW \text{ of } C_6H_{12}O_6 = 180 \text{ gms}$

$$n = \frac{wt}{G.M.wt} = \frac{540}{180} = 3 \text{ moles.}$$

(V₁₁) The empirical formula of compound is CH_2O . Its molecular weight is 90. Calculate the molecular formula of compound? (2M)

A) $Molar \text{ weight of } CH_2O = 30$

$$n = \frac{m.wt}{E.F.wt} = \frac{90}{30} = 3$$

$$Molecular \text{ formula} = (CH_2O)_3 = C_3H_6O_3$$

(V₁₂) What volume of CO_2 is obtained at STP by heating 4g of $CaCO_3$? (2M)

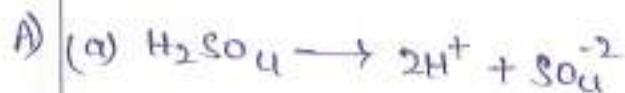
A) $100 \text{ gm of } CaCO_3 \rightarrow 22.4 \text{ lit of } CO_2$

$$4 \text{ gm} = ?$$

$$\frac{4 \times 22.4}{100} = 0.896 \text{ lit of } CO_2$$

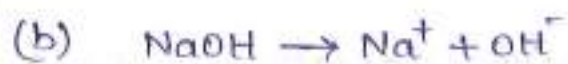
(V₁₃) Calculate the Equivalent weights of the following.

(a) H_2SO_4 (b) $NaOH$ (c) Na_2CO_3 (d) $K_2Cr_2O_7$ (2M)

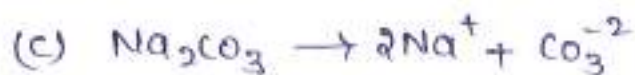


(35)

Equivalent weight of $\text{H}_2\text{SO}_4 = \frac{98}{2} = 49$



Equivalent weight of $\text{NaOH} = \frac{40}{1} = 40$.



Equivalent weight of $\text{Na}_2\text{CO}_3 = \frac{106}{2} = 53$



Equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7 = \frac{294}{2} = 147$

(V₁₄) How many numbers of CaCO_3 moles present in 200 grams of CaCO_3 ? (2M)

A) $\frac{\text{Wt}}{\text{Gm. m. wt}} = \frac{200}{100} = 2 \text{ moles}$

(V₁₅) calculate the oxidation number of 'Cr' in $\text{K}_2\text{Cr}_2\text{O}_7$ (2M)

A) $\text{K}_2\text{Cr}_2\text{O}_7 \therefore 2 \times 1 + 2 \times (2 + 7) = 0$

$$2x = 12$$

$$\boxed{x = 6}$$

$$\text{Cr}_2\text{O}_7^{2-} = 2x + (-2) \times 7 = -2$$

$$2x = 12$$

$$\boxed{x = 6}$$

(V₁₆) calculate the oxidation number of Manganese (Mn) in MnO_4^- (2M)

A) $\text{MnO}_4^- = x + 4(-2) = -1$

$$x - 8 = -1$$

$$x = 7$$

(V17) What are intensive and Extensive properties. (2M)

A) Intensive properties: The properties which are independent on the amount of substance. ex: Density

Extensive properties: The properties which are dependent on the amount of substance. ex: Mass.

(V18) Enthalpy of combustion of carbon to CO_2 is $-393.5 \text{ kJ mol}^{-1}$ calculate the heat released upon formation of 35.2 g of CO_2 from carbon and dioxygen gas. (2M)

A) $44 \text{ g of } \text{CO}_2 \Rightarrow -393.5 \text{ kJ mol}^{-1}$

$35.2 \text{ g of } \text{CO}_2 \Rightarrow ?$

$$\frac{35.2 \times -393.5}{44} = -314.8 \text{ kJ mol}^{-1}$$

(V19) What is allotropy?

A) ~~Exist~~

(V19) Write completely the Electronic configuration of K and Rb. (2M)

A) $\text{K} = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

$\text{Rb} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$

(V20) Describe the importance of plaster of paris (or) what is plaster of paris? Write its uses?

- A) 1) plaster of paris is use of building industry (2M)
2) used as bone fracture or sprain

(V₂₁) Write the average composition of portland cement (2M)

- A) $\text{CaO} = 50-60\%$, $\text{SiO}_2 = 20-25\%$, $\text{Al}_2\text{O}_3 = 5-10\%$, $\text{MgO} = 2-3\%$,
 $\text{Fe}_2\text{O}_3 = 1-2\%$ and $\text{SO}_3 = 1-2\%$.

(V₂₂) Why are alkali metals not found in the free state in nature. (2M)

- A) large size and low ionization enthalpy

(V₂₃) Which is called milk of magnesia? Give its uses? (2M)

- A) Milk of magnesia is called $\text{Mg}(\text{OH})_2$, it is used as ant acid

(V₂₄) Why are IA group elements called as alkali metals? (2M)

- A) IA group elements form hydroxides on reaction with water which are strongly alkaline

(V₂₅) Why is gypsum added to cement? (2M)

- A) To slow down the process of setting the cement so it is hardened.

(V₂₆) What does potassium carbonate can't be prepared in solway process? (2M)

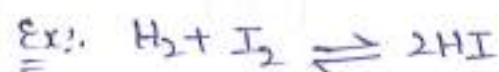
- A) The manufacture of K_2CO_3 because KHCO_3 is more soluble in water (NaHCO_3) so it can't be isolated.

(V₂₇) Write any two uses of mg metal? (2M)

- A) 1) Magnesium forms alloys with Al, Zn, Mg & tin
2) Magnesium carbonate is an ingredient of tooth paste.

(V₂₈) What is homogenous Equilibrium? Write two homogenous reaction. (2M)

- A) physical states of reactants and products are same.



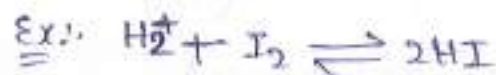
(V₂₉) What is heterogenous Equilibrium? Write two heterogenous reaction? (2M)

- A) physical states of reactants and products are different



(V₃₀) Write the relation between K_p & K_c with Example. (2M)

- A) $\Delta n = 0$ then $K_p = K_c$



(V₃₁) Given two chemical Equilibrium reactions for which $K_p = K_c$ (2M)

- A) $H_2 + I_2 \rightleftharpoons 2HI$



(V₃₂) What are Lewis acid and base? Give suitable Example (2M)

- A) To Accept lone pair of electrons is Lewis base and to

donate lone pair of electrons is called lewis base.

Ex: lewis acid $\rightarrow H^+$, BF_3

lewis base $\rightarrow NH_3$

(V₃₃) What do you mean by autoprotolysis? Give the equation to respect the autoprotolysis of water. (2M)

A) Water undergoes self ionization to form H_3O^+ and OH^- ions



(V₃₄) Why H_2O has higher B.P than H_2S . (2M)

A) Due to higher electronegativity of oxygen in H_2O

(V₃₅) Define the term hydride? (2M)

A) When dinitrogen reacts with p-block elements to form hydride

(V₃₆) How do you explain higher stability of $TlCl$ than $TlCl_3$ (2M)

A) Tl^{+1} is more stable than Tl^{+3} due to inert pair effect

(V₃₇) Explain inert pair effect? (2M)

A) The reluctance of ns_2 electrons to take part in bond formation.

(V₃₈) Give the formulae of colemanite. (2M)

A) colemanite is $Ca_2B_6O_{11} \cdot 5H_2O$.